

AJAY KUMAR GARG ENGINEERING COLLEGE, GHAZIABAD

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Sessional Test

Program: B.Tech
 Session: 2025-26
 Subject: Operating System
 Max. Marks: 50

Semester: IV
 Section: ALL CSE, IT
 Subject Code: BCS402
 Time: 2 Hours

OBE Remarks:

Q.No	1	2	3	4	5	6	7	8	9	10	11	12
CO	CO1	CO1	CO2	CO2	CO1	CO1	CO2	CO2	CO2	CO1	CO1	CO2
BL	L1	L2	L1	L2	L2	L3	L3	L2	L4	L2	L3	L5
Weightage CO1: 26.5						Weightage CO2: 23.5						

*Bloom's Level: L1: Remember, L2: Understand, L3: Apply, L4: Analyze, L5: Evaluate, L6: Create

Note: Answer all the sections with all the questions

Section-A

(2*5=10)

1. What is the role of a command interpreter in improving user convenience?
2. Differentiate between monolithic kernel and microkernel in terms of reliability.
3. Define deadlock. Can a system with a single process experience deadlock?
4. How does short-term scheduling affect system responsiveness?
5. Why is context switching considered overhead? What system component performs it?

Section-B

(5*5=25)

6. An operating system is called a resource manager and control program. Justify this statement with at least two real system scenarios.
7. Lightweight processes are preferred in modern multicore systems. Compare them with traditional processes and justify their impact on performance.
8. The performance of a CPU scheduling algorithm is measured using multiple criteria. If a scheduler minimizes turnaround time but increases response time, analyze whether it is suitable for interactive systems.
9. a) In a multilevel queue scheduler, different queues have different time quanta. Analyze how increasing the time quantum of lower-priority queues affects: Throughput and Response time
 (b) A process and a thread both represent units of execution. In which type of application would threads provide a clear advantage and why?
10. Consider the following processes for Round Robin scheduling with Time Quantum = 3 ms,
 - (i) Calculate average waiting time and turnaround time.
 - (ii) If the time quantum is increased to 5 ms, predict the change in performance.
 - (iii) Which quantum gives better response time? Justify.

Process	Arrival Time	Burst Time
P1	0	8
P2	1	4
P3	2	9
P4	3	5
P5	4	2

Section-C

(7.5*2=15)

11. Draw and explain the process state transition diagram.
 A system designer removes the "suspended" state to simplify the model.
 Evaluate how this decision will affect multiprogramming and memory utilization.
12. Consider the following snapshot of a system:

Allocation

Process	A	B	C
P0	1	2	1
P1	2	0	1
P2	3	1	2
P3	2	1	0
P4	0	2	2

Max

Process	A	B	C
P0	3	3	2
P1	4	2	2
P2	5	2	3
P3	3	3	2
P4	2	3	3

Available

A=3, B=2, C=2

- (i) Compute the **Need matrix**.
 (ii) Determine whether the system is in a **safe state**.
 (iii) If process P1 requests (1, 0, 1), should the request be granted?
 (iv) Analyze how the Banker's algorithm helps in avoiding deadlock but reduces system throughput.



Faculty Sign



HoD Sign